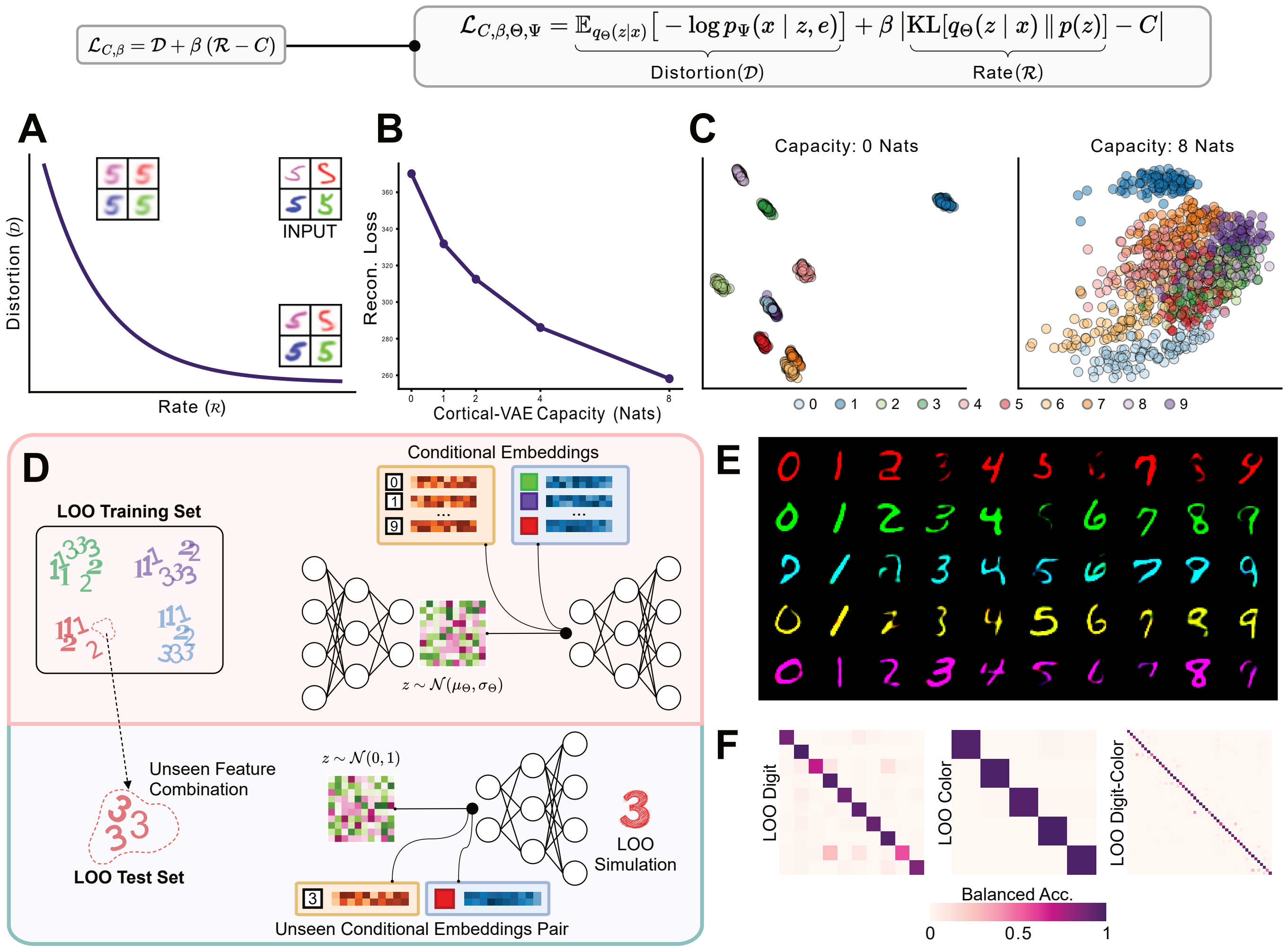

$\mathcal{L}_{C,\beta} = \mathcal{D} + \beta\,(\mathcal{R} - C)$
$\mathcal{L}_{C,\beta,\Theta,\Psi} = \underbrace{\mathbb{E}_{q_\Theta(z|x)}\left[-\log p_\Psi(x \mid z, e)\right]}_{\text{Distortion}(\mathcal{D})} + \beta\left|\underbrace{\mathrm{KL}[q_\Theta(z \mid x)\,\|\,p(z)]}_{\text{Rate}(\mathcal{R})} - C\right|$
A
Distortion ($\mathcal{D}$)
Rate ($\mathcal{R}$)
INPUT
B
Recon. Loss
Cortical-VAE Capacity (Nats)
C
Capacity: 0 Nats
Capacity: 8 Nats
0 1 2 3 4 5 6 7 8 9
D
Conditional Embeddings
LOO Training Set
$z \sim \mathcal{N}(\mu_\Theta, \sigma_\Theta)$
$z \sim \mathcal{N}(0, 1)$
Unseen Feature Combination
LOO Test Set
LOO Simulation
Unseen Conditional Embeddings Pair
E
F
LOO Digit
LOO Color
LOO Digit-Color
Balanced Acc.
0
0.5
1